

Can Vision-Language Models Perceive "Lookalike Effects" Between Humans and Cartoons as We Do?

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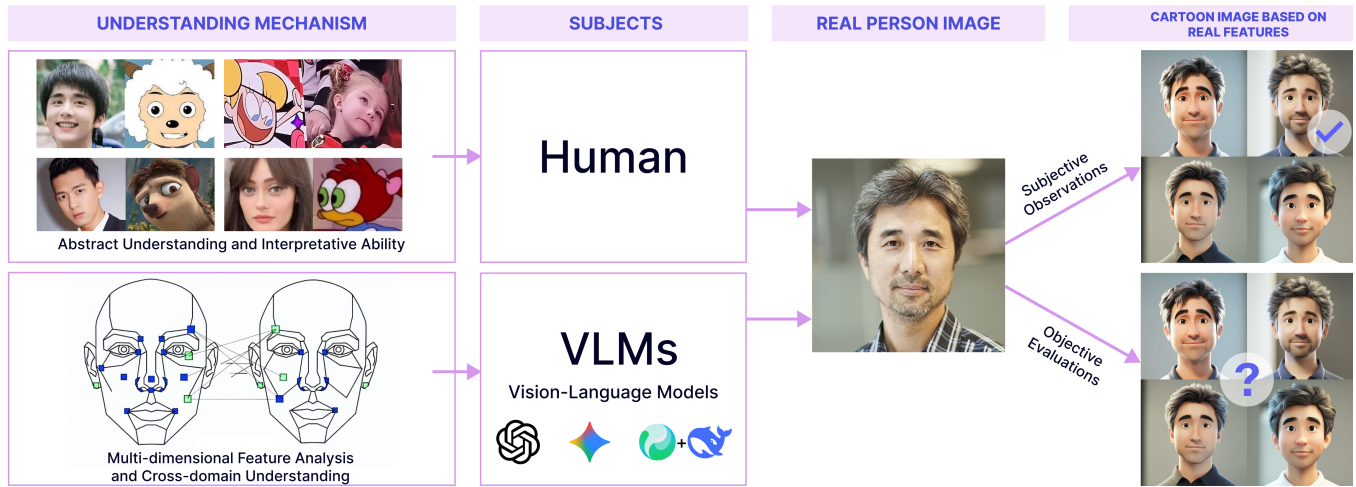


Figure 1: By comparing the perspectives of VLMs and humans on the resemblance between cartoon avatars and real people, we can understand the cognitive differences between AI systems and human visual perception.

CCS Concepts

• Computing methodologies → Visual content-based indexing and retrieval; Image manipulation.

Keywords

image understanding, resemblance perception, cartoon avatars

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1 Introduction

In the era of social media, "lookalike memes" have evolved into a unique cultural phenomenon—users are passionate about discovering and sharing striking resemblances between real people and cartoon characters (see 1, top left). These memes resonate widely because they tap into a fundamental human visual cognitive ability: our ability to capture and recognize abstract resemblant features across different visual domains [Grill-Spector and Kanwisher 2005]. This seemingly simple cognitive skill significantly surpasses current computer vision systems, which typically perform poorly when processing cross-domain resemblances, especially when involving artistic, stylized, or exaggerated forms of expression [Laishram et al. 2024].

The emergence of large Vision-Language Models (VLMs) in recent years has provided a new perspective for studying this cognitive gap. These models, trained on large-scale multimodal data, demonstrate unprecedented image comprehension capabilities. However, a key question remains: to what extent can these AI systems simulate humans' intuitive understanding of the "doppelganger" phenomenon? Can VLMs, like humans, capture subtle resemblances in facial features?

This research aims to explore the capability boundaries of contemporary VLMs in understanding resemblances between real people and cartoon characters. We have constructed a paired dataset containing photographs of real people and corresponding cartoon images in the same style, and designed a series of experiments to evaluate AI systems' ability to recognize and interpret these resemblances. By comparing AI and human judgments, we will be able to better understand the cognitive limitations of current AI systems while providing insights for developing more human-like visual understanding and interpretative capabilities in artificial intelligence.

2 Methodology

2.1 Dataset Construction

The dataset for this study includes 36 real-person portraits with diverse age and ethnicity, along with AI-generated hyper-realistic faces to ensure comprehensive sampling. For each portrait, we used Midjourney V6 to generate four cartoon images with highly consistent artistic styles through unified text prompts. This method ensures that in subsequent evaluations, variables primarily focus on capturing the "resemblance" rather than stylistic differences.

2.2 Human Subjective Benchmarks

We collected 32 human subjective evaluations as the gold standard through online questionnaires. For each real-person portrait, the questionnaire presented four corresponding cartoon images and required participants to perform a forced-choice single-selection task—choosing the one they considered "most resembling" the real person. This design effectively reduced participants' decision fatigue while directly capturing clear human consensus on optimal perceptual resemblance. Ultimately, we collected a distribution of human preferences for each sample set.

2.3 VLM Objective Evaluation

We selected three cutting-edge Vision-Language Models (VLMs) as objective evaluators: Tencent's Yuanbao multimodal model powered by deepseek, Google Gemini 2.5, and OpenAI GPT-5. To ensure the objectivity and reproducibility of the evaluation and minimize hallucination rates, we implemented a rigorous experimental protocol:

Stateless evaluation: Each sample evaluation was conducted in a new session to eliminate the model's memory effect.

Standardized input: We designed unified evaluation dimensions for all models (including facial contours, facial features, color reproduction, expression preservation with annotations) and unified text prompts.

Structured output: We used Markdown templates to standardize the model's output format for easier data parsing.

Qualitative reasonable explanations: In addition to scoring, we required models to provide brief analyses of strengths, weaknesses, or deficiencies to verify that their ratings were based on specific observations of image features rather than random generation.

3 Results & Discussion

Through our evaluation process, each set of four cartoon images received a structured evaluation table containing multi-dimensional details. Each sample underwent at least two independent evaluations, with the average total score taken as the objective evaluation result. Our study found that GPT-5, Gemini 2.5, and Yuanbao achieved hit rates of 38.9%, 52.8%, and 72.2% respectively. Notably, in the 18 samples with high human consensus (voting proportion $\geq 2/3$), the hit rates reached 55.6%, 55.6%, and 94% respectively.

Pearson correlation analysis across all 36 sample sets revealed that Yuanbao demonstrated the strongest association with human preference distribution (average correlation $r=0.46$), followed by Gemini 2.5 ($r=0.33$), with GPT-5 showing the weakest correlation ($r=0.11$). Notably, even the best-performing model showed only moderate correlation strength, indicating significant differences between AI judgment logic and human perception.

These results suggest that VLMs can identify the "optimal" cartoon images with high human consensus, but when dealing with ambiguous options where human opinions diverge, the judgment capabilities of the three models differ significantly. Our further analysis revealed several factors and phenomena that may influence model performance:

Sensitivity to image quality and pose: The accuracy of model judgment is affected by non-ideal factors in images. When real-person portraits have lens distortion, non-frontal facial orientation, or occlusive accessories such as glasses, the variability in model scoring increases, indicating that their visual analysis modules are relatively sensitive to these variables.

Potential data preferences and cultural alignment: We identified a preliminary trend in our testing: Gemini 2.5 performed better when processing non-East Asian facial samples, while Yuanbao performed more robustly on East Asian faces. Since all subjective evaluation data came from China, this suggests that model accuracy may be associated with the population distribution and cultural background in their training data. Notably, on the only sample using AI-generated hyper-realistic faces, all VLMs demonstrated remarkable consistency, with their judgment concentration even exceeding human voting.

Guidance of interactive prompts on model performance: Yuanbao's robust performance can be partially attributed to our adopted "interactive bias-correction prompting" strategy. When the model's preliminary analysis showed logical contradictions, we would prompt self-correction through a series of neutral, guiding follow-up questions (for example, "Please compare the eye sizes of the characters in the image and reassess the resemblance"). This approach significantly improved the model's logical consistency and the accuracy of its final judgment, revealing an effective way to enhance VLM performance in complex subjective evaluation tasks.

Future optimization directions: Based on feedback provided by the models in defect analysis, although they output advantages and disadvantages across different dimensions, most answers were ambiguous and lacked reference value for later modifications. Future evaluations could benefit from introducing learnable dimension weights and refined key feature dimensions—such as micro-features

like eyebrow-eye distance and eye corner morphology that may decisively influence 'resemblance' judgments. Such refinements could better align model evaluations with human perceptual priorities, thereby improving accuracy.

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